Technical Drawing & Component Specifications

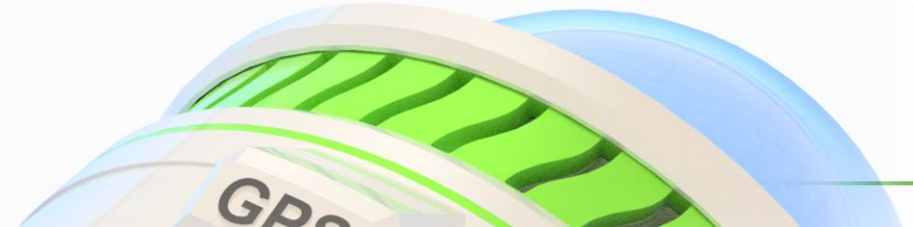
V2.0

Dec 2025



Contents

Schematic – Pin Summary 2
Schematic - Hardware & Sensor Connections 4
Schematic - Power Harness 5
Schematic - Button Harness 6
Camera & USB Peripherals 7
Base Functions 7
- COMPONENTS 9
- CS-Mount Lens 10
- Characteristic 10
- HQ Camera Mount 10
- Acrylic Dome 11
- Jack plug (female) - Optional 12
- Power Switch (3-9v) 13
- Momentary Switch (3-9v 13
- Waterproof USB-C Pass-thru 14
- Ethernet Port Pass-Thru 14
- SHT31 Humidity and Temperature Sensor 15
- GT-U7 GPS Module with Antenna 15
- M3 Threaded Inserts 16
- Piezo Buzzer 17
- RTC External Module 17



Allsky GO

Schematic – Pin Summary

HUMIDITY SHT31

Desc.	Colour
BLACK	VCC
BROWN	GND
RED	SCL
ORANGE	SDA

DS3231 (OUT)

Desc.	Colour
BROWN	GND
BLACK	VCC
ORANGE	SDA
RED	SCL

RTC DS3231 (IN)

Colour	Desc.	PIN	GPIO
ORANGE	SCL	5	**3**
YELLOW	SDA	3	**2**
GREEN	VCC	2BIS	**5V**
BLUE	**GND**	9	**GND**

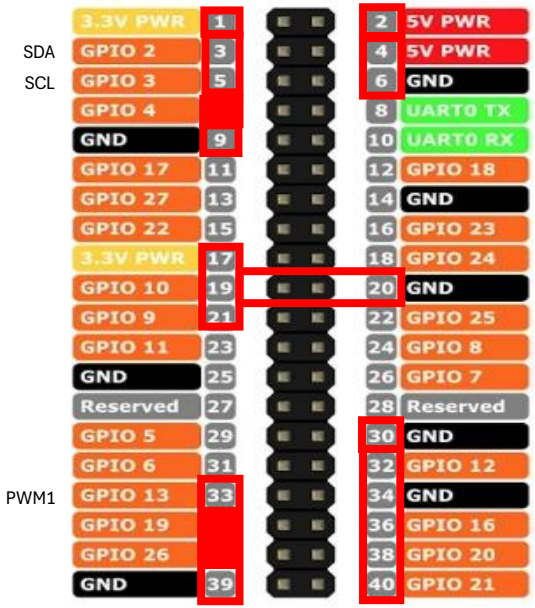
MOMENTARY BUTTON A&B

Colour	Desc.	PIN	GPIO
PURPLE	VCC	17	**3V**
WHITE	**GND**	20	
GREY	SIGNAL A	21	**9**
BLUE	SIGNAL B	19	**10**

PIEZO BUZZER

Colour	Desc.	PIN	GPIO
RED	PWM1	33	**13**
		35	
		37	
BLACK	**GND**	39	**GND**

RASPBERRY PI4 _ ALLSKYGO



CPU FAN

GPIO	PIN	Desc.	Colour
5V	4	VCC	RED
GND	6	**GND**	BLACK

POWER BUTTON

GPIO	PIN	Desc.	Colour
3V	1	VCC	RED
GND	30	**GND**	BLACK

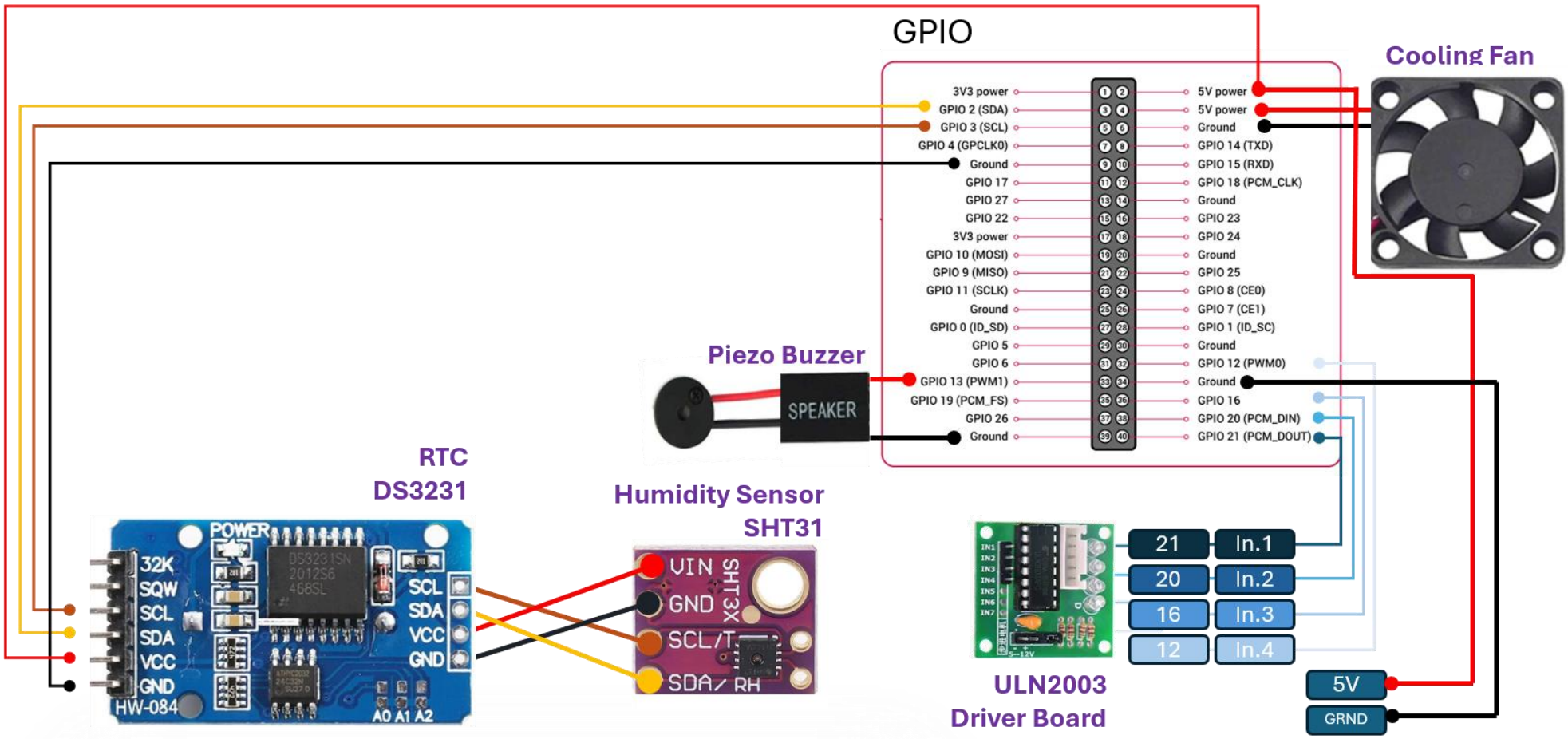
ULN2003 STEPPER MOTOR DRIVER

GPIO	PIN	Desc.	Colour
5V	2BIS	VCC	ORANGE
GND	34	**GND**	YELLOW
21	40	IN1	WHITE
20	39	IN2	BROWN
16	36	IN3	BLACK
12	32	IN4	RED



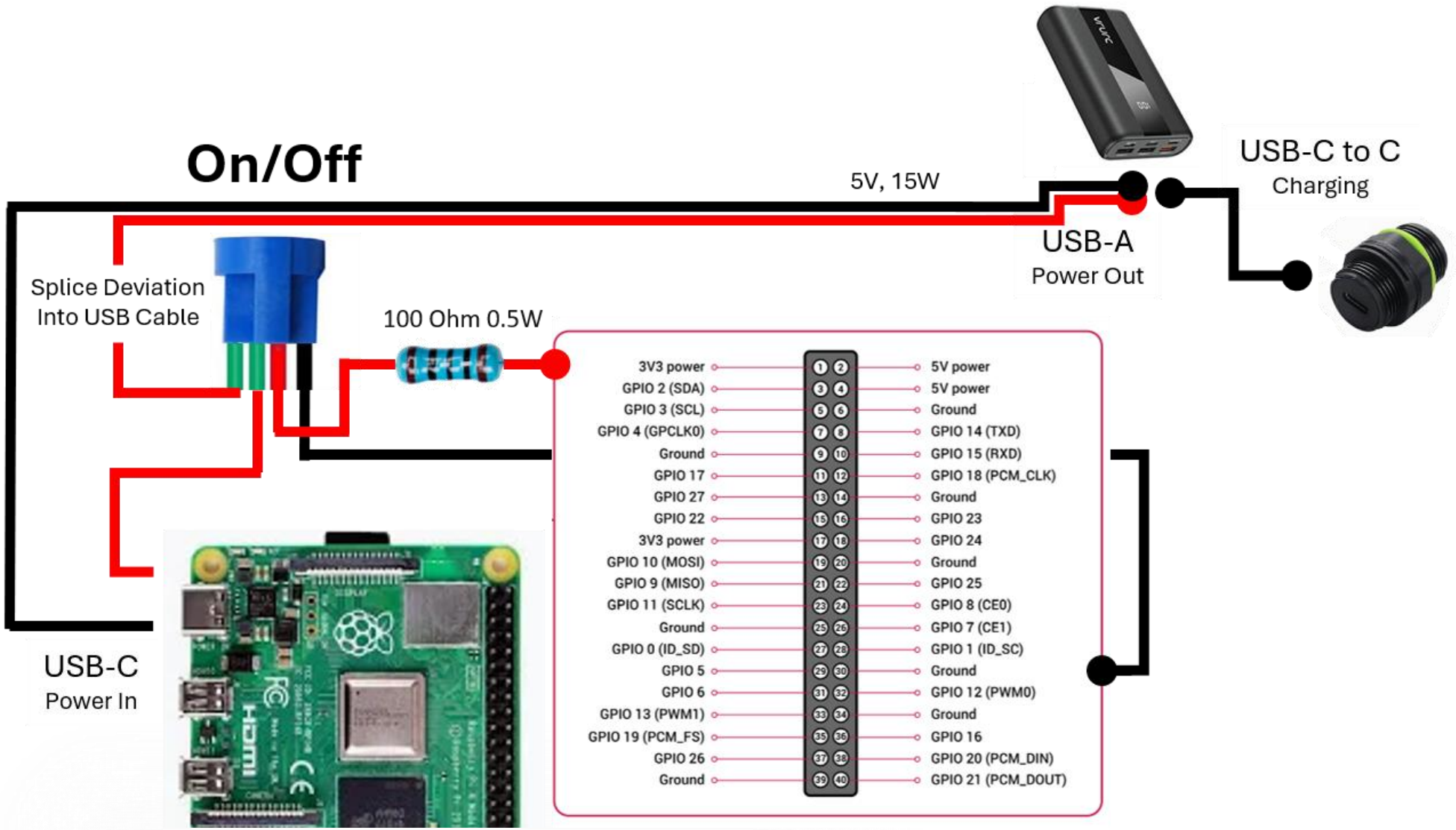
Allsky GO

Schematic - Hardware & Sensor Connections



Allsky GO

Schematic - Power Harness



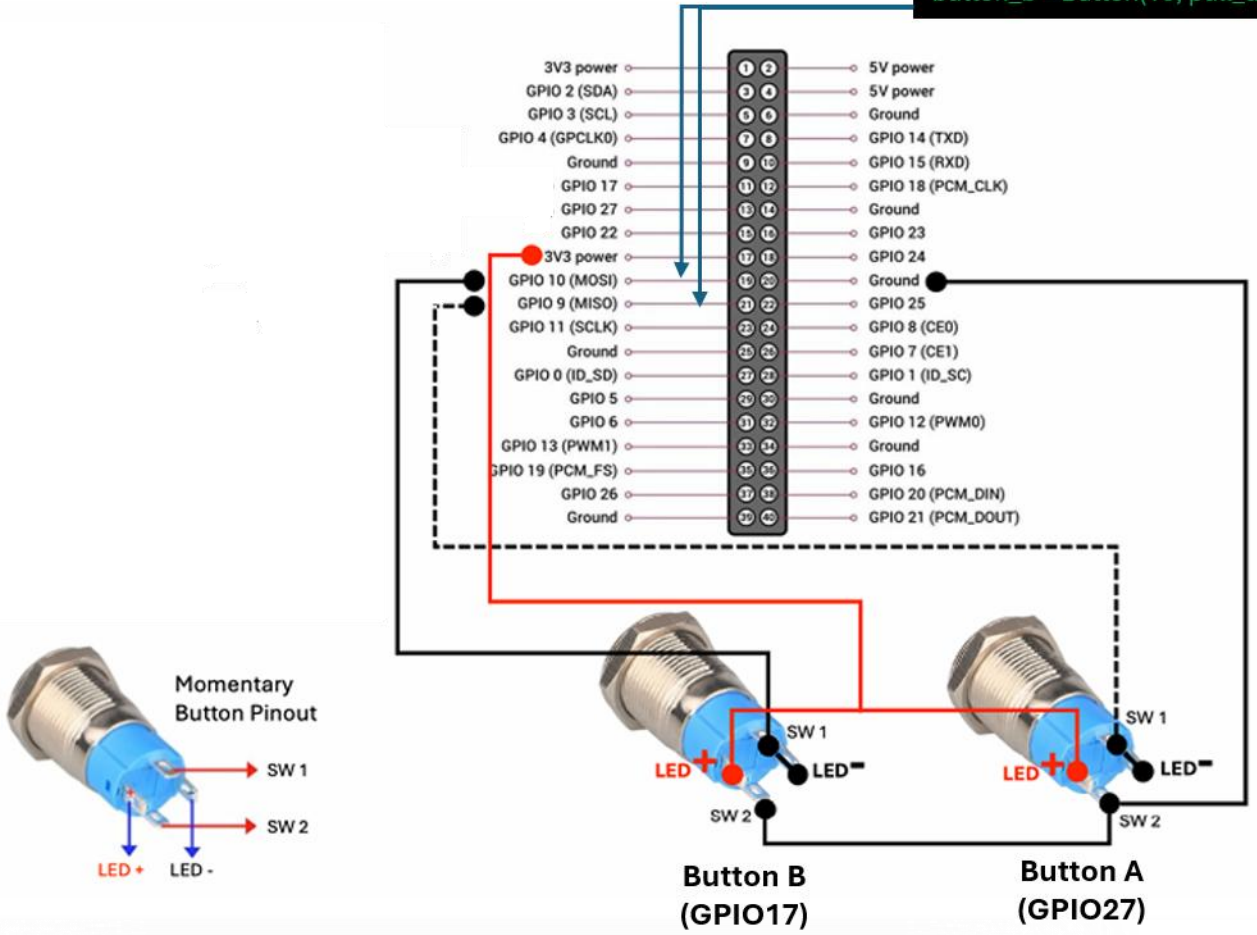
Allsky GO

Schematic - Button Harness

Schematics:

Internal Pull-up Resistors Enabled

```
button_a = Button(9, pull_up=True)  # Button A - Use pull-up mode
button_b = Button(10, pull_up=True)  # Button B - Use pull-up mode
```

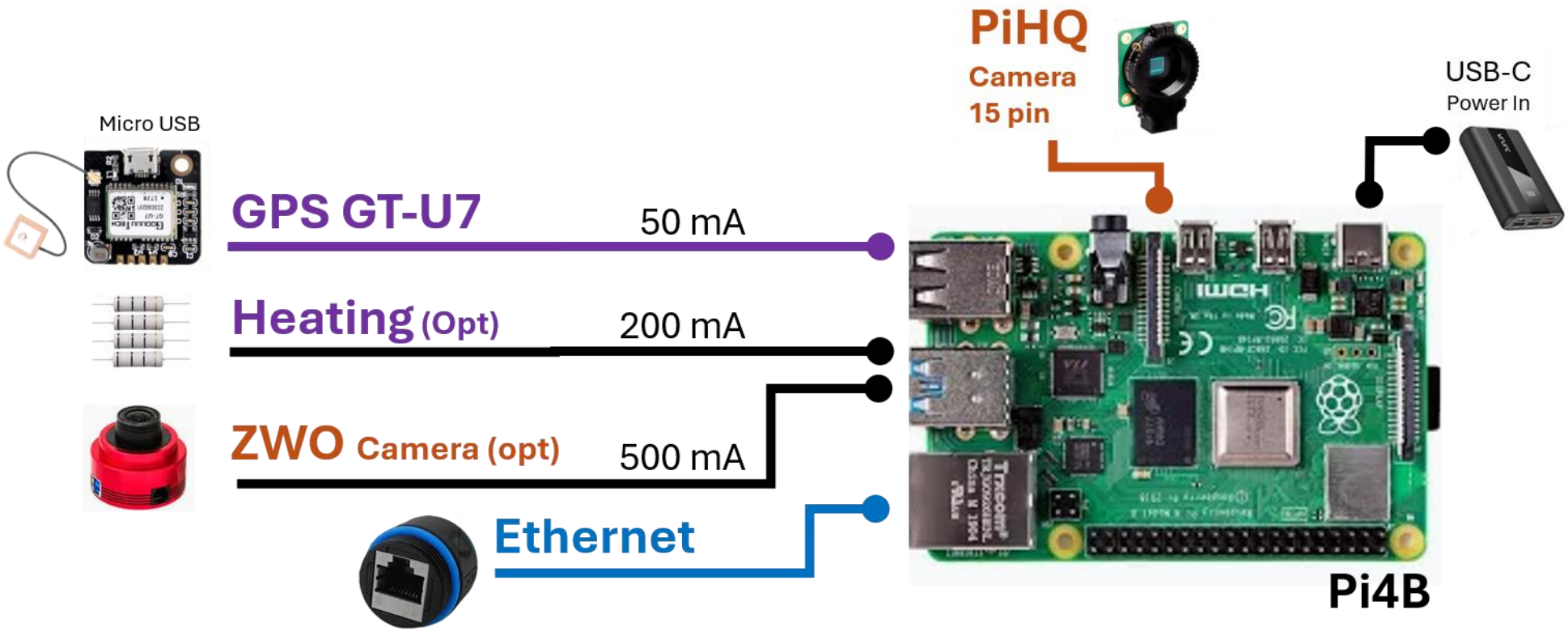


Thanks to
@Aemideus_3927449



Allsky GO

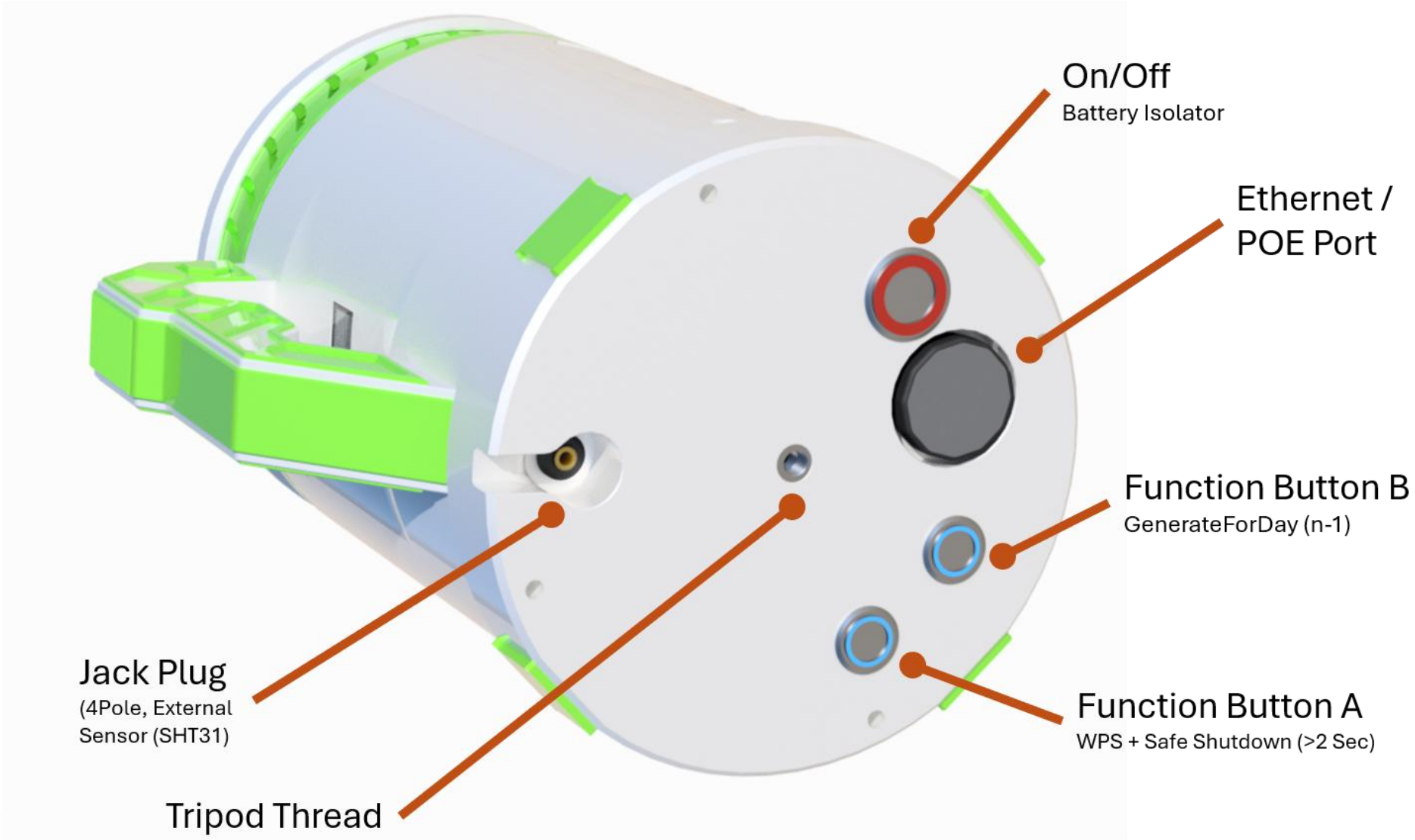
Camera & USB Peripherals





Allsky GO

Base Functions





Allsky GO

COMPONENTS

CS-Mount Lens

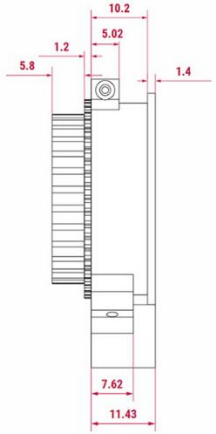
Characteristic 2.5 or 2.8mm for 150-170deg FOV [AliExpress]

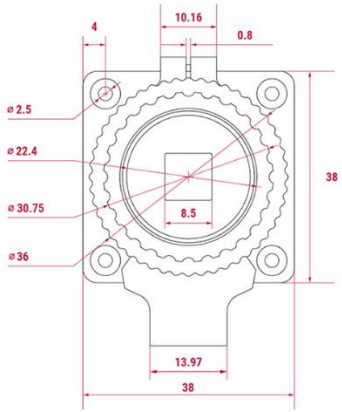


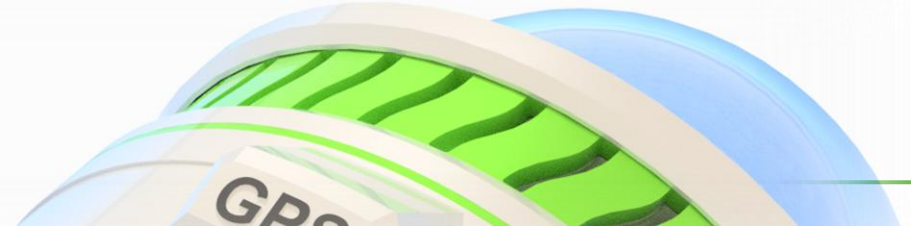
HQ Camera Mount

Mark: Raspberry [Reichelt]

[CS-Mount_HQ_Camera_Mechanical_Drawings]





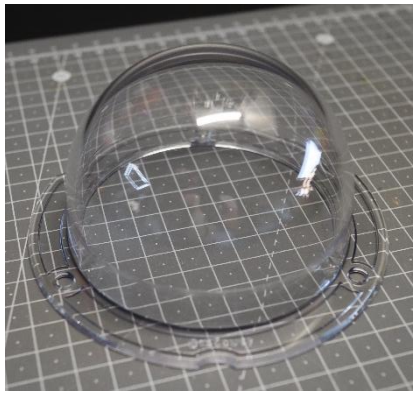


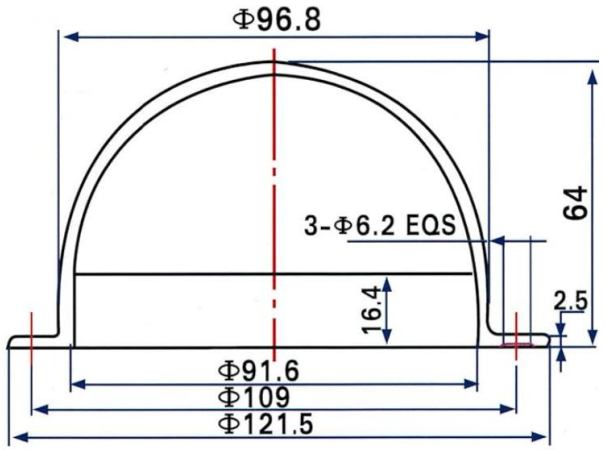
Allsky GO

Acrylic Dome

Brand: JMC Amazon

Characteristic 3.5Inc Dome





Power Bank

Make: Vrurc Amazon (VRURC Mini Power bank)

Characteristic: 20.000mah



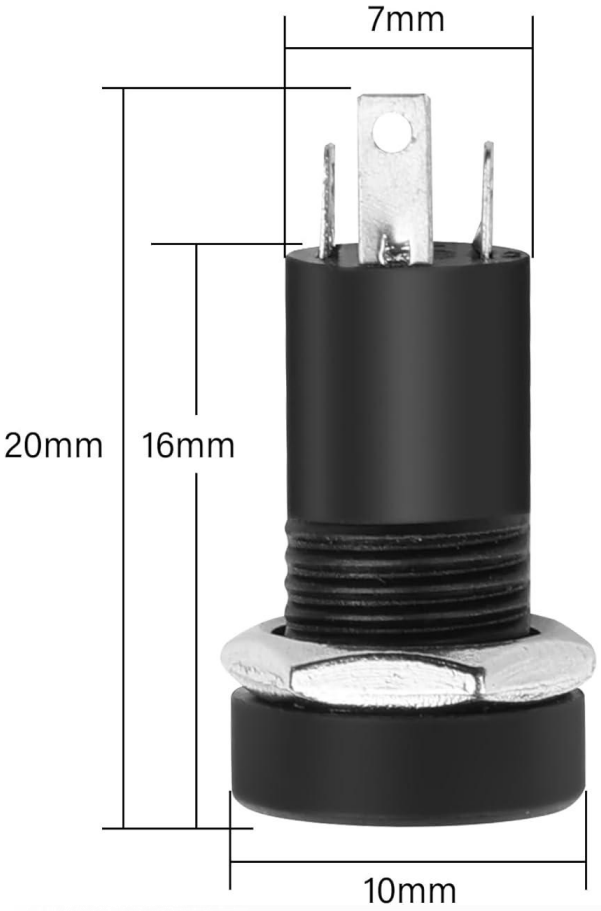


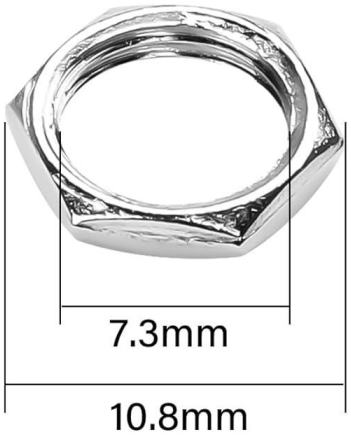
Allsky GO

Jack plug (female) - Optional

Distributor: PNGKNYOCN - Amazon

Characteristic: 4 Pole – Requires 4 Pole Jack Plug

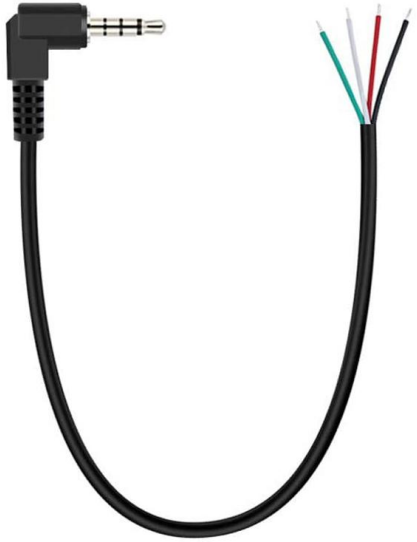




Jack plug (male)

90degree – 4 Pole – Length 30cm

Amazon





Allsky GO

Power Switch (3-9v)

Distributor: Gebildet Amazon

Characteristic: 16mm Thread

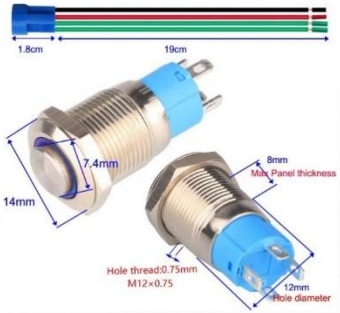


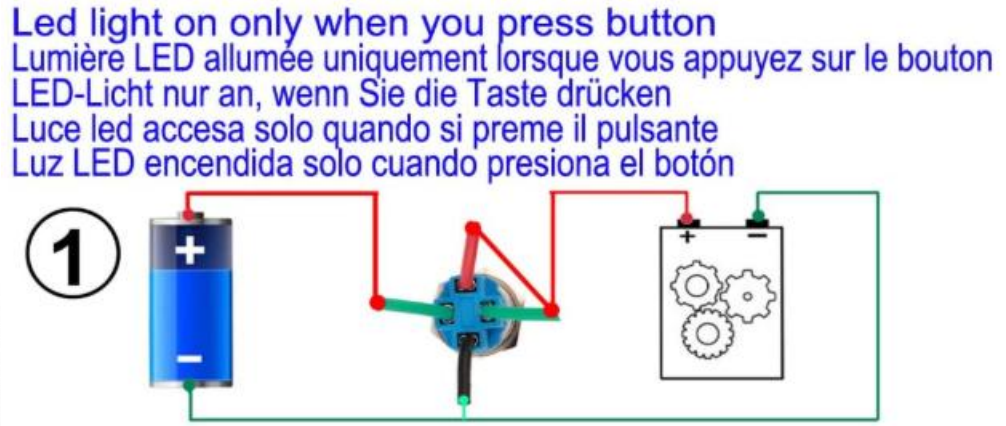


Momentary Switch (3-9v)

Distributor: Gebildet Amazon

Characteristic: 12mm Thread







Allsky GO

Waterproof USB-C Pass-thru (only the core section required)

Distributor: PNGKNYOCN Amazon

Characteristic: 19mm Thread



Ethernet Port Pass-Thru

Distributor: Greluma Amazon

Characteristic: 24mm Diameter



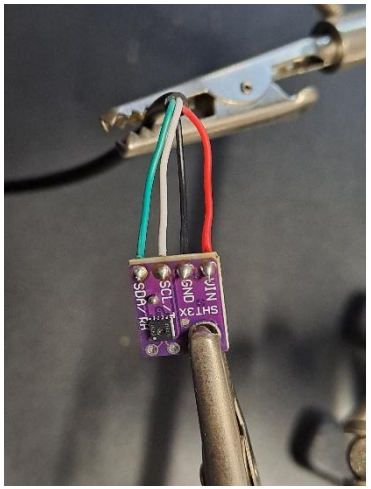


Allsky GO

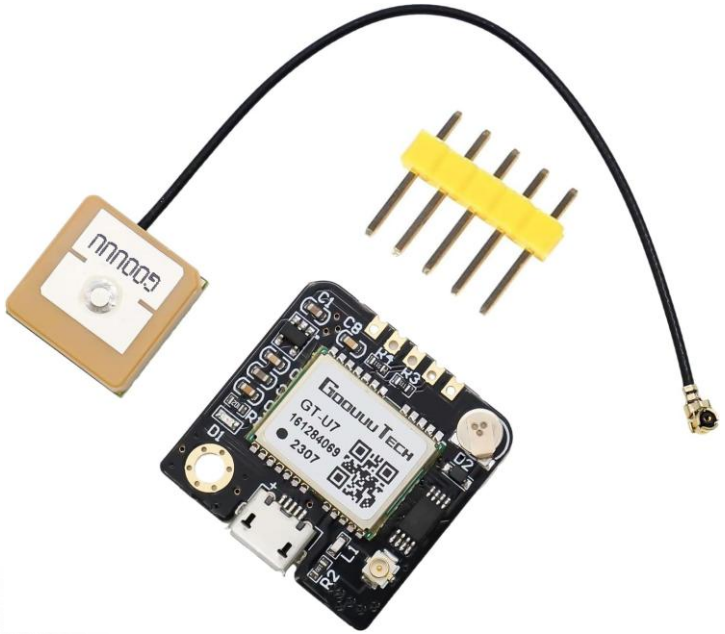
SHT31 Humidity and Temperature Sensor

Different Options exist depending on your circumstance. Sensors with 4 poles can be wired to the Jack Plug and programmed for Raspberry.

Amazon







GT-U7 GPS Module with Antenna

Amazon – Goouu Tech



Allsky GO

¼-20 Tripod Threaded Inserts

Mark: Ruthex Amazon –

Characteristic: 7.9-8mm hole – two lengths available (longer prefered)



M3 Threaded Inserts

Characteristic: 3.9-4.0mm width

Mark: Ruthex Amazon



Both Insert Types available in the same set : AMAZON



Allsky GO

Metal Oxide Film Resistors

Distributor: PATIKIL Amazon

Characteristic: 100 Ohm 5W

Amazon.it



Piezo Buzzer

Distributor: ICQUANZX Amazon

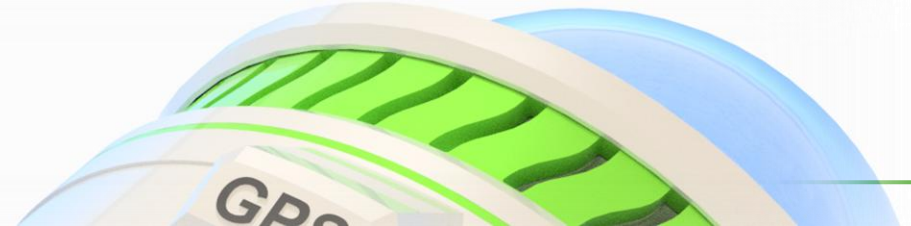


RTC External Module

Distributor: AZ Delivery Amazon

Characteristics: Coin Battery CR2032





Allsky GO